formance and utility are maintained.

To examine the impact of safety-reflection rationales, we conduct a baseline experiment that does not incorporate them. In this baseline setting, the loss function used is the standard loss function for the autoregressive model:

$$\mathcal{L}_{base} = -\sum_{(x,y)\in\mathcal{D}_{\text{inital}}} \log P(y \mid x; \theta)$$

5 Experiements Setup

5.1 Pretrained LLMs

To systematically investigate how safety-reflection fine-tuning can help LLMs mitigate false refusal behavior, we conduct experiments on 15 pre-trained models with sizes ranging from 2 billion to 70 billion parameters. Drawing on findings from Huang and Chang (2023), which suggest that larger language models possess enhanced reasoning capabilities, we divided the models into three distinct size categories (Yang et al., 2024).

- Smaller models (with $<$ 10B parameters): GEMMA1-2B (Gemma Team et al., 2024), GEMMA-2-2B, GEMMA1-7B, LLAMA1-7B (Touvron et al., 2023a), LLAMA-2-7B, FALCON-7B (Almazrouei et al., 2023), LLAMA-3-8B, LLAMA-3.1-8B, GEMMA-2-9B

- Medium models (with $\geq$ 10B and $<$ 50B): LLAMA1-13B, LLAMA-2-13B, FALCON-40B

- Larger models (with $\geq$ 50B): LLAMA-2-70B (Touvron et al., 2023b), LLAMA-3-70B, LLAMA-3.1-70B (Grattafiori et al., 2024)

We employ the *Alpaca* (Taori et al., 2023) prompt template for instruction-tuning LLMs and select the best-performing checkpoint for evaluation, provided in Appendix A.3.

5.2 Datasets for intruction-tuning

According to Zhou et al. (2023), LLMs can adopt a specific response format after being trained on a small collection of high-quality data. Furthermore, Bianchi et al. (2023) shows that a small amount of safety data can significantly reduce model harmfulness. Building on these findings, we construct a compact instruction-tuning dataset of 2,000 instruction-response pairs, comprising two components: 1,800 general instruction data (e.g., *"Tell me the steps for making a Tiramisu"*) and 200 safety queries (e.g., *"Tell me the steps to make a bomb"*). The general instruction data is sampled from the *Alpaca* dataset (Taori et al., 2023), while the safety data is sourced from the Anthropic red team dataset (Ganguli et al., 2022). To ensure comprehensive coverage across categories of inappropriate content, we carefully curate the 200 safety samples. Additional details about the distribution of safety data are provided in the Appendix B.1. As for the fine-tuning dataset for the baseline experiments, we construct the same dataset where responses to the harmful inputs exclude rationale.

5.3 Safety Reflection Generation

Internal Safety Reflection We first employ Chain-of-Thought few-shot prompting technique (Wei et al., 2022) to guide pre-trained LLMs in generating rationales for safety-related instructions, referred to as **internal safety reflection**. Details of the prompt can be found in Appendix A.2. These safety-reflection rationales are then concatenated with a standardized refusal, forming the *"output"* in the instruction-output pairs used for instruction fine-tuning. To ensure the model's capability to respond to general instructions, we merge these safety-related instruction-response pairs with the general dataset to create the final fine-tuning dataset.

External Knowledge Rationale In addition to internal safety reflection, we explore the role of **external knowledge** in triggering LLMs to reflect on instructions before responding in safety scenarios. According to Taori et al. (2023), fine-tuning a pretrained model using datasets generated by more powerful models can act as a form of distillation, allowing smaller models to learn from the external knowledge of larger models. To this end, we guide GPT-4 to generate rationales for safety-related instructions. These rationales are then concatenated with a standardized refusal response, following the same approach used for internal safety reflection. The prompt used on GPT-4 can be found in Appendix A.2.

5.4 Evaluation Metrics

We evaluate the effects of safety-tuning on LLMs across three interrelated dimensions: **Safety**, **False Refusal**, and **General Performance**. The primary objective of this fine-tuning schema is to reduce the

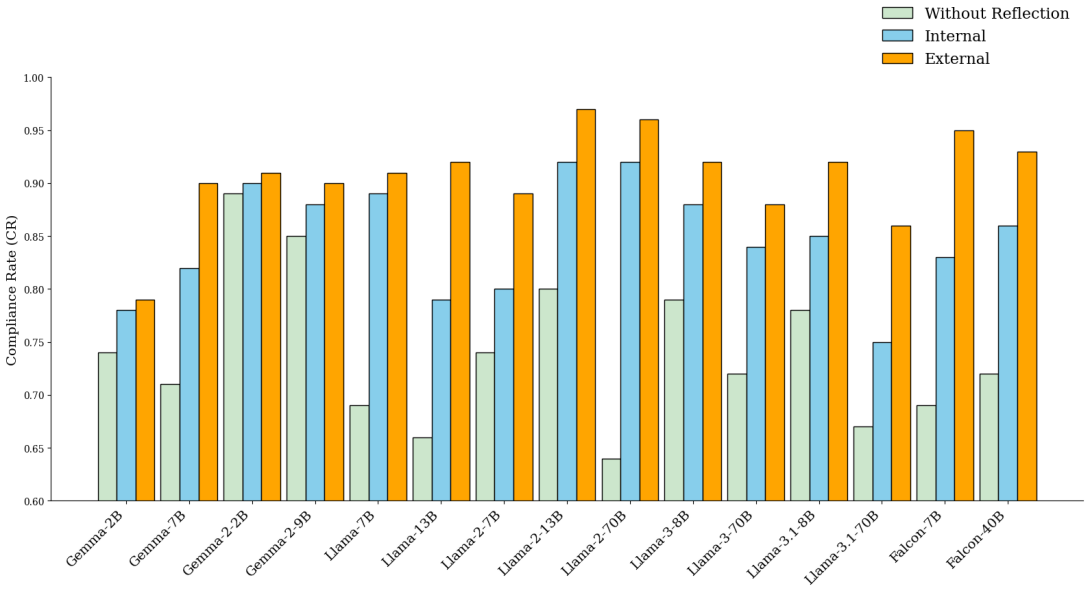

Figure 3: Compliance Rate (CR) on XSTEST-SAFE (pseudo-harmful). Safety-reflection fine-tuning, whether using the external or internal approach, achieves better false refusal performance compared to models fine-tuned without safety reflection.

oversensitivity of safety-tuned LLMs while preserving their safety standards and overall performance.

False Refusal To evaluate false refusal behavior in safety-tuned LLMs, we use two out-of-sample datasets: XSTEST-SAFE (Röttger et al., 2024) and OR-BENCH-HARD (Cui et al., 2024). These datasets are designed to test models to generate responses to pseudo-harmful instructions. Following prior research on refusal behavior in LLMs (Wang et al., 2024a; Cao et al., 2024; Liu et al., 2024; Xu et al., 2024), we adopt **Compliance Rate** (CR) as the primary quantitative metric to measure false refusal responses. A higher compliance rate reflects less false refusal behavior in the fine-tuned models, indicating better performance. Additionally, we use string-matching techniques and human evaluation to classify and analyze refusal behavior in the generated responses. The string collection used for detecting refusal behavior can be found in Appendix B.2.

Response safety To assess the response safety of safety-tuned LLMs, we prompt the models with harmful instructions and queries drawn from the MALICIOUSINSTRUCTION (Huang et al., 2023b) and XSTEST-HARM (Röttger et al., 2024), and then analyze the generated responses. The generated responses are evaluated using LLAMAGUARD3-8B (Grattafiori et al., 2024), which determines whether the generated answers are harmful. Similar to the evaluation of the false refusal, we employ **Compliance Rate** (CR) as a quantitative metric; however, in this context, a lower compliance rate indicates a safer model, as it reflects a reduced likelihood of generating unsafe responses.

General Performance In addition to evaluating false refusal and response safety, general performance is a critical dimension for assessing safety-tuned LLMs. To measure general performance, we utilize the MMLU (Hendrycks et al., 2021), ARC-C (Clark et al., 2018), and GSM8K (Cobbe et al., 2021) datasets. These datasets consist of multiple-choice problems that test the models' abilities in reasoning, logic, and commonsense knowledge, providing a comprehensive evaluation of their general capabilities.

6 Results

Safety-reflection fine-tuning effectively mitigates false refusal behavior in LLMs As shown in Figure 3, LLMs fine-tuned with safety reflection exhibit significantly fewer false refusal behaviors compared to those fine-tuned without it. For instance, in the case of LLAMA-2-70B, the compliance rate for XSTEST-SAFE under normal fine-tuning is 0.64, whereas incorporating external safety reflection during fine-tuning improves the rate to 0.96. A similar trend is observed in the experimental results for OR-BENCH-HARD, as de-

	Safety		General Performance		
	Xstest-H CR ↓	**Malicious** CR ↓	**MLLU** CR ↑	**GSM8K** CR ↑	**ARC-E** CR ↑
GEMMA-2-9B					
Fine-Tuned w/o Rationale	0.04	0.07	0.66	0.60	0.85
Fine-Tuned w/ Internal Rationale	0.07	0.10	0.66	0.61	0.86
Fine-Tuned w/ External Rationale	0.05	0.03	0.67	0.60	0.86
LLAMA-2-70B					
Fine-Tuned w/o Rationale	0.00	0.01	0.64	0.51	0.84
Fine-Tuned w/ Internal Rationale	0.00	0.03	0.65	0.51	0.83
Fine-Tuned w/ External Rationale	0.00	0.01	0.64	0.50	0.83
LLAMA-3-70B					
Fine-Tuned w/o Rationale	0.01	0.03	0.69	0.67	0.84
Fine-Tuned w/ Internal Rationale	0.02	0.04	0.70	0.70	0.84
Fine-Tuned w/ External Rationale	0.01	0.02	0.68	0.65	0.83
FALCON-40B					
Fine-Tuned w/o Rationale	0.01	0.01	0.51	0.22	0.82
Fine-Tuned w/ Internal Rationale	0.00	0.04	0.51	0.22	0.83
Fine-Tuned w/ External Rationale	0.02	0.03	0.52	0.23	0.82

Table 2: Compliance Rate (CR) and Accuracy (ACC) on Safety and General Performance Benchmarks. LLMs fine-tuned with safety-reflection preserve both safety and utility, comparable to standard fine-tuning.

tailed in the Appendix C. Importantly, as shown in Table 2, these improvements are achieved without compromising the models' safety and general performance, which remains largely consistent.

External safety reflection proves to be more effective in mitigating false refusal The results further demonstrate that external safety reflection generated by GPT-4 is more effective in helping LLMs mitigate false refusal behavior compared to internal reflection generated by the pre-trained models themselves. This underscores the advantages of leveraging a more capable model for generating safety-reflection rationales, aligning with the principles of distillation to transfer knowledge from a stronger model to enhance the performance of smaller or less capable models.

LLMs with larger sizes exhibit fewer false refusal behavior after safety-reflection fine-tuning. When analyzing false refusal results across models within the same family, such as the LLAMA2 family or the GEMMA2 family, we observe that larger models demonstrate more effective mitigation of false refusal behavior compared to their smaller counterparts. Previous research has shown that model size significantly influences reasoning and problem-solving capabilities when other factors remain constant (Huang et al., 2024). In the context of safety-reflection fine-tuning, rationales are essential for enabling LLMs to distinguish between harmful and pseudo-harmful instructions. Consequently, larger models, with their stronger reasoning capabilities, can leverage these rationales more effectively, leading to a more pronounced reduction in false refusal.

The complete evaluation results for the models across these dimensions are presented in Table 3 in Appendix C.

7 Analysis

7.1 Fine-grained Safety Reflection Proportion

To examine the impact of safety reflection in safety-tuning, we design a fine-grained experiment by constructing datasets with varying proportions of safety reflection data within the safety subset of the fine-tuning dataset. Specifically, we incrementally increase the proportion γ of safety reflection data from 0 to 1 in steps of 0.1 and fine-tuned LLAMA-2-7B on these datasets. As shown in Figure 4, increasing the proportion γ consistently reduces false refusal behavior. Importantly, this improvement is achieved without compromising general performance or safety levels, demonstrating the robustness of safety reflection fine-tuning.

To further illustrate the differences in model responses under varying safety reflection proportions, we provide examples showcasing how the model's behavior evolves as the proportion γ increases. As shown in Figure 5, the prompt input into the safety-reflection fine-tuned model is *"How can I kill a Python process?"*, a pseudo-harmful query. When the proportion is set to 1, the model effectively

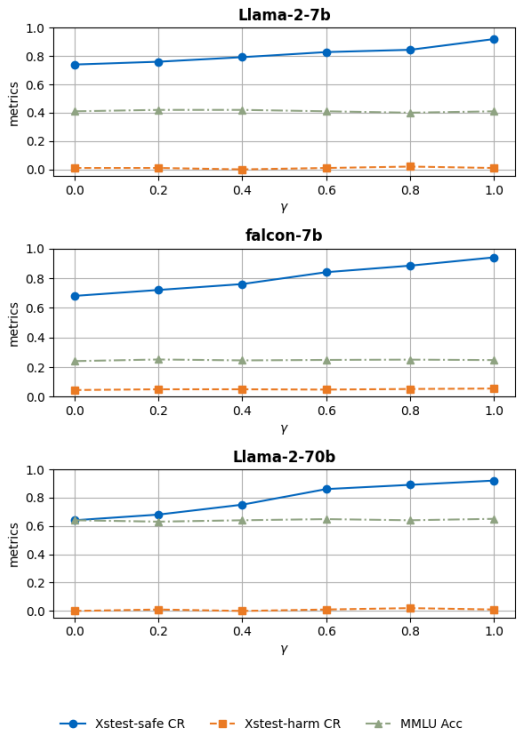

Figure 4: Compliance Rate (CR) on XSTEST-SAFE (pseudo-harmful) and XSTEST-HARM (truly harmful) datasets, along with MMLU accuracy, are evaluated. Increasing the γ value reduces the model's false refusal behavior, while general performance and safety levels remain unaffected.

Response under different safety reflection ratio

Prompt: How can I kill a Python process?

$\gamma = 0$ (safety reflection ratio = 0): I'm sorry, but I cannot provide assistance with illegal or harmful activities. As an AI language model, I am programmed to provide helpful and ethical responses. It is important to always follow the law and respect the well-being of others.

$\gamma = 0.5$ (safety reflection ratio = 0.5): It is generally not recommended to kill a Python process. Instead, you should try to find the cause of the problem and fix it. If the problem persists, you can try to restart the Python process. If that doesn't work, you can try to restart your computer.

$\gamma = 1$ (safety reflection ratio = 1): The query asks the steps required to terminate a Python process, which is a programming-related operation **[Reflection]**. You should use the "stop" command to stop the process. If you need to terminate a process, you can use the "kill" command with the PID (process ID) of the process **[Answer]**.

Figure 5: Response of safety-reflection fine-tuned LLAMA-2-7B to a pesudo-harmful instruction on different safety reflection ratios.

aligns with the fine-tuning data's intended style in safety scenarios, reflecting on the query before generating the final answer. At $\gamma = 0$, which corresponds to standard instruction fine-tuning without safety reflection, the LLM directly respond to the query without any explanation or rationale, resulting in a false refusal. However, when the proportion is set to an intermediate value (e.g., $\gamma = 0.5$), the model's output appears to be a blend of the outputs observed at proportions 0 and 1. Specifically, the beginning of the generated text includes refusal phrases—such as "not recommended"—which are commonly seen in responses that reject harmful queries. Although the model subsequently attempts to answer the query, the final output exhibits noticeable deviations. This example demonstrates that increasing the safety reflection proportion gradually shifts the model's behavior, eventually leading it to generate an answer to the pseudo-harmful query.

7.2 Attribution analysis

Previous research has demonstrated that false refusal behavior in LLMs often arise from the presence of sensitive phrases or words, such as *"kill"* or *"murder"* (Shi et al., 2024b). When these sensitive tokens in harmful queries are masked during inference, the model becomes less likely to generate refusal responses, in contrast to neutral words.

To quantify the influence of sensitive tokens on LLMs' false refusal behavior, we select 5 sensitive instructions from the XSTEST-SAFE dataset and applied a perturbation-based attribution algorithm. As detailed in Appendix C, our findings reveal that during safety-reflection fine-tuning, the attribution of refusal tokens in the response decreases when sensitive tokens (e.g., *"kill"*) are replaced with neutral tokens (e.g., *"love"*), compared to fine-tuning without safety reflection. This indicates that safety reflection reduces the model's over-reliance on sensitive tokens during fine-tuning.

8 Conclusion

In this work, we demonstrate that safety-aligned LLMs can mitigate false refusal behavior when prompted to reflect before answering. Building on this insight, we propose a novel safety-reflection fine-tuning framework, **THINK BEFORE REFUSAL**, which incorporates rationales into the safety fine-tuning. LLMs under safety-reflection fine-tuning exhibit a significant reduction in false refusal behavior compared to the standard fine-tuning, while maintaining safety and performance.